OOPS ASSIGNMENT-8

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Function Template

Q1.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <typename T>
void swapValues(T &a, T &b) {
    T temp;
    temp = a;
    a = b;
    b = temp;
}

int main() {
    cout<<"Puneet Singh, 1024150037\n";
    int x, y;
    cout << "Enter two integers: ";
    cin >> x >> y;

    cout << "Before swapping: " << x << " " << y << endl;
    swapValues(x, y);
    cout << "After swapping: " << x << " " << y << endl;

    float a, b;
    cout << "Enter two float numbers: ";
    cin >> a >> b;

    cout << "Before swapping: " << a << " " << b << endl;
    swapValues(a, b);
    cout << "After swapping: " << a << " " << b << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Enter two integers: 84
36
Before swapping: 84 36
After swapping: 36 84
Enter two float numbers: 23.11
96.49
Before swapping: 23.11 96.49
After swapping: 96.49 23.11


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <typename T>
T findMin(T arr[], int size) {
    T min = arr[0];   // assume first element is minimum

    for(int i = 1; i < size; i++) {
        if(arr[i] < min) {
            min = arr[i];
        }
    }
    return min;
}

int main() {
    cout<<"Puneet Singh, 1024150037\n";
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int arr[100];
    cout << "Enter elements: ";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    cout << "Minimum element is: " << findMin(arr, n) << endl;

    // Example with float
    float farr[3] = {2.5, 1.2, 3.8};
    cout << "Minimum in float array: " << findMin(farr, 3) << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Enter number of elements: 4
Enter elements: 45
456
56
48
Minimum element is: 45
Minimum in float array: 1.2


=== Code Execution Successful ===
```

Q3.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <typename T>
void bubbleSort(T arr[], int size) {
    for(int i = 0; i < size - 1; i++) {
        for(int j = 0; j < size - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                // swap
                T temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main() {
    cout<<"Puneet Singh, 1024150037\n";
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int arr[100];
    cout << "Enter elements: ";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    bubbleSort(arr, n);

    cout << "Sorted array: ";
    for(int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;

    // Example with float
    float farr[5] = {3.2, 1.5, 4.8, 2.1, 0.9};
    bubbleSort(farr, 5);

    cout << "Sorted float array: ";
    for(int i = 0; i < 5; i++) {
```

Output

```
Puneet Singh, 1024150037
Enter number of elements: 4
Enter elements: 14
1456
44

12
Sorted array: 12 14 44 1456
Sorted float array: 0.9 1.5 2.1 3.2 4.8

=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <typename T>
int linearSearch(T arr[], int size, T key) {
    for(int i = 0; i < size; i++) {
        if(arr[i] == key) {
            return i;   // return index if found
        }
    }
    return -1;  // not found
}

int main() {
    cout<<"Puneet Singh, 1024150037\n";
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int arr[100];
    cout << "Enter elements: ";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    int key;
    cout << "Enter element to search: ";
    cin >> key;

    int result = linearSearch(arr, n, key);

    if(result != -1)
        cout << "Element found at index: " << result << endl;
    else
        cout << "Element not found" << endl;

    // Example with float
    float farr[5] = {1.1, 2.2, 3.3, 4.4, 5.5};
    float fkey = 3.3;

    int res2 = linearSearch(farr, 5, fkey);
    if(res2 != -1)
        cout << "Float element found at index: " << res2 << endl;
```

Output

```
Puneet Singh, 1024150037
Enter number of elements: 4
Enter elements:
56
84
29
43
Enter element to search: 84
Element found at index: 1
Float element found at index: 2


=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;

// 1. Single parameter
template <typename T>
void process(T a) {
    cout << "Single parameter: " << a << endl;
}

// 2. Two parameters of same type
template <typename T>
void process(T a, T b) {
    cout << "Two parameters (same type): " << a << " and " << b << endl;
}

// 3. Two parameters of different types
template <typename T1, typename T2>
void process(T1 a, T2 b) {
    cout << "Two parameters (different types): " << a << " and " << b << endl;
}

int main() {
    cout<<"Puneet Singh, 1024150037\n";
    process(10);              // single parameter
    process(5, 15);          // same type (int, int)
    process(3.5, 2.1);       // same type (float, float)
    process(10, 3.14);       // different types (int, float)
    process('A', 100);       // different types (char, int)

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Single parameter: 10
Two parameters (same type): 5 and 15
Two parameters (same type): 3.5 and 2.1
Two parameters (different types): 10 and 3.14
Two parameters (different types): A and 100


=== Code Execution Successful ===
```

Class Template

Q1.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <class T>
class Stack {
    T arr[100];
    int top;

public:
    Stack() { top = -1; }

    void push(T value) {
        if(top == 99) {
            cout << "Stack Overflow\n";
            return;
        }
        arr[++top] = value;
    }

    void pop() {
        if(top == -1) {
            cout << "Stack Underflow\n";
            return;
        }
        cout << "Popped: " << arr[top--] << endl;
    }
};

int main() {
    cout<<"Puneet Singh, 1024150037\n";

    Stack<int> s;
    s.push(10);
    s.push(20);
    s.pop();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Popped: 20


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <class T>
class Queue {
    T arr[100];
    int front, rear;

public:
    Queue() {
        front = -1;
        rear = -1;
    }

    void enqueue(T value) {
        if(rear == 99) {
            cout << "Queue Overflow\n";
            return;
        }
        if(front == -1) front = 0;
        arr[++rear] = value;
    }

    void dequeue() {
        if(front == -1 || front > rear) {
            cout << "Queue Underflow\n";
            return;
        }
        cout << "Dequeued: " << arr[front++] << endl;
    }
};

int main() {
    cout<<"Puneet Singh, 1024150037\n";

    Queue<int> q;
    q.enqueue(5);
    q.enqueue(10);
    q.dequeue();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Dequeued: 5


=== Code Execution Successful ===
```

Q3.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <class T1, class T2>
class Pair {
    T1 a;
    T2 b;

public:
    Pair(T1 x, T2 y) {
        a = x;
        b = y;
    }

    void display() {
        cout << "Pair: " << a << ", " << b << endl;
    }
};

int main() {
    cout<<"Puneet Singh, 1024150037\n";

    Pair<int, float> p1(10, 5.5);
    p1.display();

    Pair<char, int> p2('A', 100);
    p2.display();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Pair: 10, 5.5
Pair: A, 100


=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <class T>
class Arithmetic {
    T a, b;

public:
    Arithmetic(T x, T y) {
        a = x;
        b = y;
    }

    void add() {
        cout << "Addition: " << a + b << endl;
    }

    void subtract() {
        cout << "Subtraction: " << a - b << endl;
    }

    void multiply() {
        cout << "Multiplication: " << a * b << endl;
    }

    void divide() {
        if(b != 0)
            cout << "Division: " << a / b << endl;
        else
            cout << "Division by zero error\n";
    }
};

int main() {
    cout<<"Puneet Singh, 1024150037\n";

    Arithmetic<int> obj(10, 5);
    obj.add();
    obj.subtract();
    obj.multiply();
    obj.divide();
```

Output

```
Puneet Singh, 1024150037
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2


=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;

template <class T>
class Array {
    T arr[100];
    int n;

public:
    void input() {
        cout << "Enter number of elements: ";
        cin >> n;

        cout << "Enter elements:\n";
        for(int i = 0; i < n; i++) {
            cin >> arr[i];
        }
    }

    void display() {
        cout << "Array elements: ";
        for(int i = 0; i < n; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    cout<<"Puneet Singh, 1024150037\n";

    Array<int> a;
    a.input();
    a.display();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Enter number of elements: 4
Enter elements:
12
56
8
23
Array elements: 12 56 8 23


=== Code Execution Successful ===
```